LIGHTYEAR

MS 107 Release safeguards and evidence reconciliation

Recorded October 15 2026. Implementation: PRs #342 through #349. This record extends MS104 and MS106.

LIGHTYEAR now checks the conditions behind a release decision before that decision can be signed or consumed. The refreshed evidence explains which INTCALC checks passed, which exclusions would change its measured score, and why the program still cannot be released. Customers and reviewers can trace the conclusion to specific source files, measurements and approvals.

Conditions and current evidence

The signing boundary verifies the four retained threshold conditions and their source hashes. A request with unmet conditions is refused with `threshold-conditions-unmet`. Historical conditional decisions remain readable; their existence does not make them effective. The follow-up adds a structural basis for a proposed I/O exception and retries transient compiler-download failures without changing the pinned checksum.

The Factory status page and decision packet derive from the latest sealed release evidence and a retained public activation snapshot. This removes stale status text while preserving the original dates of measurements and decisions. The snapshot describes the existing local signed-only installation; these merges do not update its pinned runtime or sign new records.

What the fresh INTCALC run established

The retained successful main campaign is run 38763552148 at commit `f343925026bee705ed3c16a734dc5bae4b24c588`. Two independent calculations agree on the projected denominators. A mutant is an intentional program change used to test whether the comparisons detect wrong behavior.

Measure	Recorded result
Business outcomes	8 of 8 covered
I/O outcomes	85 of 86 covered; one structural exception remains proposed
Raw mutation result	308 detected changes out of 421 sites
Loader gaps reassessed	113 of 113 resolved by valid comparisons
Projected twin mutation result	308 of 308 after the applicable exclusions are accepted
Projected z/OS-facing result	308 of 421
Constant output fields	Seven fields each have one source-assigned value

The 113 original gap records are retained. Reassessing them removes no mutation sites and grants no extra detection or coverage credit. The projected twin score is conditional; it is not the current signed release result. Explicit scope records now control whether an exclusion applies to the twin or to a z/OS-facing claim. Missing or mismatched scope fails closed, and English wording no longer selects the denominator.

Proposed policy and guarded execution

Threshold/2 requires two observed values per output field. Seven source-assigned constants cannot meet that requirement. Howard selected preparation of threshold/3 with explicit constant-field exemptions. The unsigned proposal binds each exemption to the source, line and mutation evidence. It exempts only the diversity requirement; the fields must still be compared and mutation-tested. Threshold/3 is not implemented or activated by these merges, and INTCALC remains unreleasable under unchanged threshold/2.

POSTTRAN Phase 2 now requires a signed proof bound to its plan, candidate, compiler image and trusted journal before reserving its one permitted attempt. The downloaded image must match its hash and have at least 24 hours of retention. Equal return codes 0 or 4 may agree; equal 8 or 12 diverge. Candidate 999 versus twin 12 is allowed only for the declared duplicate-ID scenario. No Phase 2 comparison was authorized or run by this delivery. Integration CI found that the retained plan predates the new scope implementation. Preparation now validates the retained files at their recorded commit and reports current drift. Execution from the integrated checkout still refuses that drift and requires a new reviewed plan.

The retained compiler artifact expires October 29 at 14:49:11 UTC. Its 24-hour retention requirement ends the current authorization window on October 28 at 14:49:11 UTC, subject to the final live retention check.

Remaining evidence and decisions

B06's signed attempt and cleanup evidence remains unavailable on this Mac. The Windows folder was not searched and no native attempt was repeated. A read-only inventory helper and a dated access-block record make that limitation explicit. Large original INTCALC receipts still await the designated retention store.

The six-request signing path requires an explicit threshold/3 implementation and a successful throwaway-authority rehearsal before real signatures. The proposed I/O exception remains unsigned. These are separate from the completed code-merge authorization.

Supporting records

- Merged PR and commit reconciliation
- Fresh INTCALC evidence
- Explicit claim scopes
- Unsigned threshold proposal
- POSTTRAN authorization bindings
- B06 evidence access record